ECE 310 Fall 2026
Lecture 6: z-transform

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Learning Objectives

- Explain the motivation for using the z-transform.
- Define the z-transform of a discrete-time signal.
- Explain the significance of the region of convergence (ROC) for a z-transform.
- Compute the z-transform by inspection using a table of transform pairs.



What have we accomplished thus far?



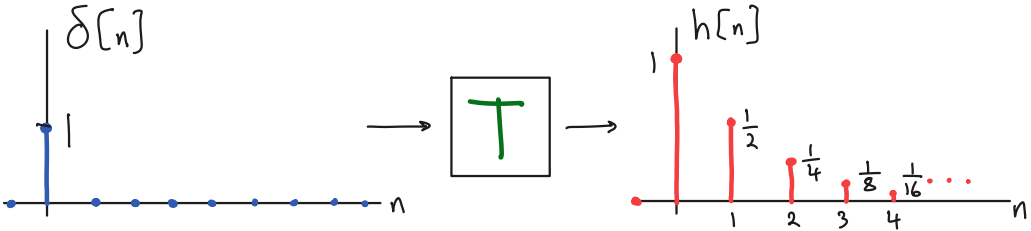
What have we accomplished thus far?

- We can find the output of any LTI system to any input signal using its impulse response $h[n]$, i.e. by performing convolution.
- We can find the impulse response of any LTI system, e.g. described by an LCCDE.



What have we accomplished thus far?

- We can find the output of any LTI system to any input signal using its impulse response $h[n]$, i.e. by performing convolution.
- We can find the impulse response of any LTI system, e.g. described by an LCCDE.
- The above points are important and highly significant! However, do they necessarily tell us what a system does?
- Consider an LTI system with $h[n] = \left(\frac{1}{2}\right)^n u[n]$.



Motivation for the z-transform

What do we want instead?: Can we find a particular class of signals that do not change shape when passing through LTI systems? They can be scaled or shifted by the system, but their shape does not change.



Motivation for the z-transform

Ex: $z_0 = 1+j2 \rightarrow x[n] = (1+j2)^n$

Consider $x[n] = z_0^n$, for any $z_0 \in \mathbb{C}$.

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If $h[n]$ is the impulse response of an LTI system,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, z_0^{n-k}$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, z_0^{n} z_0^{-k}$$

$$= z_0^n \sum_{k=-\infty}^{\infty} h[k]\, z_0^{-k}$$

$$= z_0^n \, H(z_0) = x[n] H(z_0)$$

$H(z_0)$ is a constant and only depends on $h[n]$ and $z_0 \in \mathbb{C}$.

↳ No dependence on n!



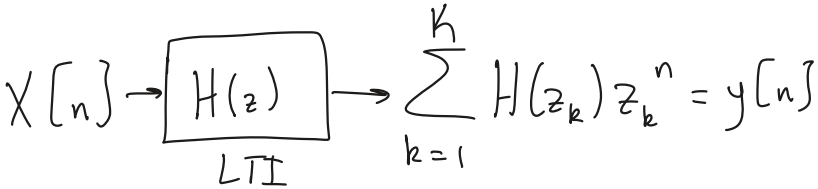
where $H(z_0) = \sum_{k=-\infty}^{\infty} h[k]\, z_0^{-k}$

is the transfer function of the system

Motivation for the z-transform

How does knowing a system's transfer function $H(z)$ help us?

If $x[n] = \sum_{k=1}^{K} z_k^n$



- $|H(z_k)| > 1$: Amplify or emphasize
- $0 < |H(z_k)| < 1$: Attenuate or weaken
- $|H(z_k)| = 0$: Filter or stop.

z-transform

The z-transform $X(z)$ of a discrete-time signal $x[n]$ is given by

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$x[n] \overset{\mathcal{Z}}{\mapsto} X(z)$$

We must always specify a *region of convergence* to make a z-transform unique!

Region of Convergence

The *region of convergence* (ROC) is the values of z where the z-transform sum converges for a given signal.



Computing the z-transform

$$x[n] = u[n]$$

$$X(z) = \sum_{n=-\infty}^{\infty} u[n] z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} u[n] z^{-n} = \sum_{n=0}^{\infty} (1) \cdot z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n$$

$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$ if $|a| < 1$, else $\rightarrow$ does not converge

$|z^{-1}| < 1$

$|\frac{1}{z}| < 1$

$\frac{1}{|z|} < 1$

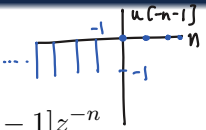
$|z| > 1$

$$= \frac{1}{1 - z^{-1}}, \quad |z^{-1}| < 1 \quad \text{or} \quad |\tfrac{1}{z}| < 1$$

$$= \frac{1}{1 - z^{-1}}, \quad \text{ROC: } |z| > 1$$

z-transform is only unique if ROC is specified!

$$x[n] = -u[-n-1]$$



$$X(z) = \sum_{n=-\infty}^{\infty} -u[-n-1] z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{-1} (-1) \cdot z^{-n} = \sum_{n=-\infty}^{-1} -z^{-n}$$

$m = -n-1$, $n = -m-1$

$$= \sum_{m=0}^{\infty} -z^{-(-m-1)} = -\sum_{m=0}^{\infty} z^{m+1}$$

$$= -z \sum_{m=0}^{\infty} z^m = -z \left(\frac{1}{1-z} \right), \quad \text{ROC: } |z| < 1$$

$$= \frac{-z}{1-z} \cdot \frac{-z^{-1}}{-z^{-1}} = \frac{1}{1 - z^{-1}}, \quad \text{ROC: } |z| < 1$$

Example: Computing the z-transform

Compute the z-transform of $x[n] = a^n u[n]$ where $a \in \mathbb{C}$.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n$$

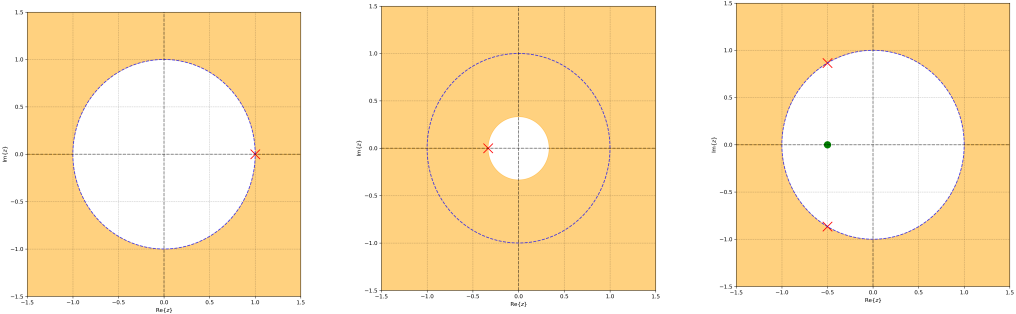
$$= \frac{1}{1 - \frac{a}{z}}, \quad \text{ROC} = \left|\frac{a}{z}\right| < 1$$

$$= \frac{1}{1 - az^{-1}}, \quad \text{ROC} = |z| > |a|$$

Pole-Zero Plots

For a z-transform $X(z)$,

- The values of z for which $X(z) = 0$ are known as *zeros*.
- The values of z for which $X(z) \to \infty$ are known as *poles*.
- *Pole-zero plots* visualize the ROC, poles, and zeros of a z-transform.



(a) $H(z) = \frac{1}{1-z^{-1}}$, $|z| > 1$ (b) $H(z) = \frac{1}{1+\frac{1}{3}z^{-1}}$ (c) $H(z) = \frac{1+\frac{1}{2}z^{-1}}{1+z^{-1}+z^{-2}}$

Figure 1: Example pole-zero plots for (a) $x[n] = u[n]$, (b) $x[n] = \left(-\frac{1}{3}\right)^n u[n]$, and (c) $\cos\left(\frac{2}{3}\pi n\right) u[n]$. The dashed blue circle represents the unit-circle.

Common z-transform Pairs

Table 1: Common z-transform pairs.

$x[n]$	$X(z)$	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
$-a^n u[-(n+1)]$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
$-na^n u[-(n+1)]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
$\sin(\omega_0 n)u[n]$	$\frac{\sin(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
$a^n \cos(\omega_0 n)u[n]$	$\frac{1-a\cos(\omega_0)z^{-1}}{1-2a\cos(\omega_0)z^{-1}+a^2z^{-2}}$	$\|z\| > \|a\|$
$a^n \sin(\omega_0 n)u[n]$	$\frac{a\sin(\omega_0)z^{-1}}{1-2a\cos(\omega_0)z^{-1}+a^2z^{-2}}$	$\|z\| > \|a\|$